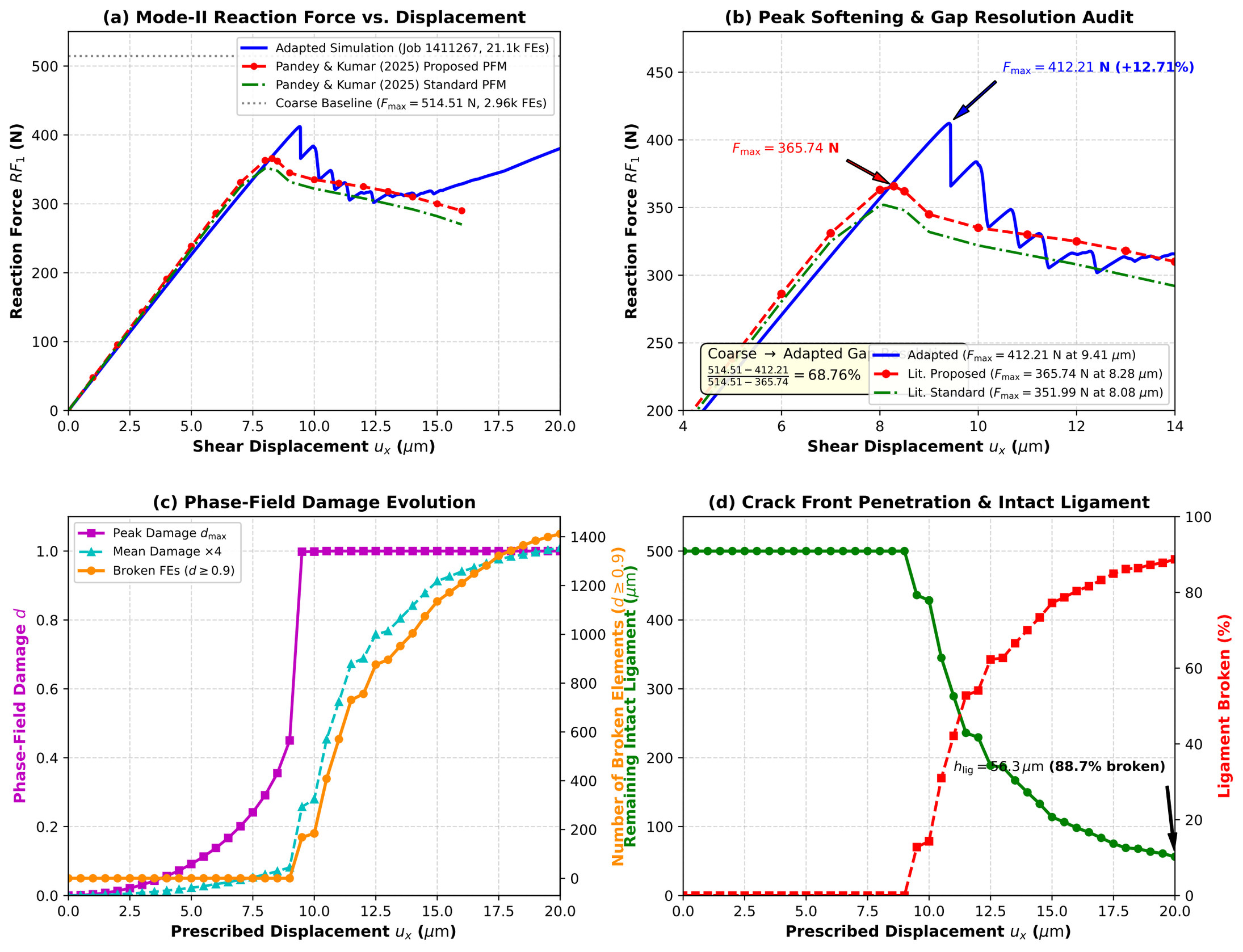
(a) Mode-II Reaction Force vs. Displacement
Adapted Simulation (Job 1411267, 21.1k FEs)
Pandey & Kumar (2025) Proposed PFM
Pandey & Kumar (2025) Standard PFM
Coarse Baseline (F_{max} = 514.51 N, 2.96k FEs)
Reaction Force RF_1 (N)
Shear Displacement u_x (μm)
(b) Peak Softening & Gap Resolution Audit
F_{max} = 412.21 N (+12.71%)
F_{max} = 365.74 N
Coarse → Adapted Ga
$\frac{514.51-412.21}{514.51-365.74}$ = 68.76%
Adapted (F_{max} = 412.21 N at 9.41 μm)
Lit. Proposed (F_{max} = 365.74 N at 8.28 μm)
Lit. Standard (F_{max} = 351.99 N at 8.08 μm)
Reaction Force RF_1 (N)
Shear Displacement u_x (μm)
(c) Phase-Field Damage Evolution
Peak Damage d_{max}
Mean Damage ×4
Broken FEs ($d \geq 0.9$)
Phase-Field Damage d
Number of Broken Elements ($d \geq 0.9$)
Remaining Intact Ligament (μm)
Prescribed Displacement u_x (μm)
(d) Crack Front Penetration & Intact Ligament
h_{lig} = 56.3 μm (88.7% broken)
Ligament Broken (%)
Prescribed Displacement u_x (μm)